THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · MEDIUM LEVEL FULL-YEAR PAPER SET 1

Full Syllabus (Chapters 1–8) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. 1,25,000 × 8 = 1

(a) 1,00,00,000 **(b)** 9,90,000 **(c)** 10,00,000 **(d)** 10,01,000

2. The value of $72 \div (4 + 2) \times (10 - 7)$ is: 1

(a) 193 **(b)** 31 **(c)** 36 **(d)** 38

3. $12.5 \times 0.08 =$ 1

(a) 100 **(b)** 0.1 **(c)** 10 **(d)** 1

4. The value of $3x + 2y - 5$ for $x = 4$ and $y = 6$ is: 1

(a) 18 **(b)** 21 **(c)** 20 **(d)** 19

5. An angle is 2 times its complement. The angle is: 1

(a) 60° **(b)** 55° **(c)** 70° **(d)** 65°

6. The smallest digit $*$ making $5 * 48$ divisible by 3 is: 1

(a) 2 **(b)** 3 **(c)** 1 **(d)** 0

7. The angles of a triangle are in the ratio $2 : 3 : 4$. The largest angle is: 1

(a) 100° **(b)** 80° **(c)** 90° **(d)** 70°

8. $1\frac{1}{2} \times 2\frac{1}{3}$ equals: 1

(a) 7 **(b)** $3\frac{1}{2}$ **(c)** $1\frac{3}{4}$ **(d)** $2\frac{1}{6}$

9. 3,12,50,000 ÷ 25 = 1

(a) 1251000 **(b)** 1249900 **(c)** 1250000 **(d)** 1250100

10. The value of $120 \div 4 \div 5 + 3 \times 7$ is: 1

(a) 63 **(b)** 25 **(c)** 27 **(d)** 29

11. $1.44 \div 1.2 =$ 1

(a) 12 **(b)** 0.12 **(c)** 1.2 **(d)** 120

12. Simplify $9x + 4y - 5x + 3y$: 1

(a) $4x + y$ **(b)** $11xy$ **(c)** $4x + 7y$ **(d)** $14x + 7y$

13. Half of the supplement of 40° is: 1

(a) 60° **(b)** 90° **(c)** 80° **(d)** 70°

14. The Collatz sequence from 8 (including 8 and 1) has how many terms? 1

(a) 6 **(b)** 3 **(c)** 4 **(d)** 5

15. A base angle of an isosceles triangle is 35°. The exterior angle at the apex is: 1

(a) 70° **(b)** 60° **(c)** 80° **(d)** 90°

16. $1\frac{1}{2} \div \frac{3}{4}$ equals: 1

(a) 2 **(b)** 1 **(c)** 4 **(d)** $1\frac{1}{8}$

17. A crore is how many times 1,000? 1

(a) 9999 **(b)** 10010 **(c)** 10001 **(d)** 10000

18. The value of $6 \times (12 - 5)$ added to 8 is: 1

(a) 60 **(b)** 49 **(c)** 51 **(d)** 50

19. The value of $2.5 \times 4 + 1.75$ is: 1

(a) 11.75 **(b)** 11.85 **(c)** 11.65 **(d)** 12.75

20. If $x = 8$, then $3(x - 2) + 5$ equals: 1

(a) 22 **(b)** 24 **(c)** 23 **(d)** 25

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. Find 2,34,000 × 15. 2

22. Simplify $60 \div (5 + (9 - 4)) \times 3$. 2

23. Simplify $5.6 + 3.45 - 2.8$. 2

24. Simplify: $3(x + 4) + 5x$. 2

25. An angle is 20° more than its complement. Find the angle. 2

26. The number $5 * 14$ is divisible by 9. Find the digit $*$ and write the number. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. A train covers 1,25,000 km in a year. How many kilometres will it cover in 8 years and how many such trains are needed to cover 60,00,000 km in 8 years? 3

28. A trader buys 50 articles at ₹40 each and sells 30 of them at ₹60 each and the rest at ₹35 each. Write an expression for the profit/loss and evaluate it. 3

29. A car runs 12.5 km on 1 litre of petrol. How far does it run on 8 litres? How many litres are needed for 75 km? 3

30. Two numbers are $7n + 5$ and $3n - 2$. Find their sum and their difference (first minus second), and evaluate both for $n = 4$. 3

31. Two lines intersect and two adjacent angles are in the ratio $1 : 4$. Find all four angles. 3

OR

Apply Kaprekar's routine to 7621 repeatedly until you reach 6174. 3

32. The angles of $\triangle ABC$ are in the ratio $2 : 3 : 4$. Find the angles and the exterior angle at the vertex of the largest angle. 3

OR

A sheet is $1\frac{1}{2}$ m long and $2\frac{1}{4}$ m wide. Find its area and perimeter. 3

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SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. A company earned ₹12,50,000 in the first year, and in each next year the earning increased by ₹1,50,000. Find its earnings in the 2nd, 3rd and 4th years and the total of the four years. **5**

34. A school buys 40 chairs at ₹450 each and 10 tables at ₹1200 each. A discount of ₹50 per chair is given. Write an expression for the amount payable and find it. **5**

35. A wire 9.5 m long is cut into pieces of 1.2 m each. How many whole pieces are made and what length is left over? **5**

36. A shopkeeper sells x kg of rice at ₹48 per kg and y kg of wheat at ₹36 per kg. Write an expression for his total sales and find it when $x = 10$ and $y = 15$. **5**

OR

Two interior angles on the same side of a transversal cutting two parallel lines are in the ratio $2 : 3$. Find both angles and the angle forming a linear pair with the larger one. **5**

37. Write the Collatz sequence from 7. Find the largest number reached, the number of even terms and the number of odd terms. **5**

OR

The equal sides of an isosceles triangle are $(2x + 1)$ cm and $(3x - 4)$ cm and its base is $(x + 3)$ cm. Find x, the three sides and the perimeter. Check that the triangle exists. **5**

38. A wall of 240 m^2 is painted in three days. On day 1, $\frac{1}{4}$ of it is painted; on day 2, $\frac{2}{3}$ of the remainder; the rest on day 3. Find the area painted on each day. **5**

OR

The distance between two cities is 12,50,000 m. A runner covers 40,000 m each day. How many full days does he run, how much distance is left after that, and how many days in all are needed to finish? **5**

END OF QUESTION PAPER — Total: 80 Marks

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